

GUNASREE R

Salem, Tamil Nadu 636003 | AI Engineer | gunasreer@gmail.com | [linkedin.com/in/gunasree-r-55024224a/](https://www.linkedin.com/in/gunasree-r-55024224a/) | <https://github.com/rgunasree> | <https://gunasree-portfolio.vercel.app>

Summary

AI Engineer specializing in LLM-powered applications, Retrieval-Augmented Generation (RAG), and data-driven AI systems. Experienced in designing scalable pipelines integrating embeddings, vector search, and PostgreSQL for real-time applications. Strong focus on building production-grade ML systems, optimizing inference performance, and translating data into actionable insights.

Achievements

- **Winner:** *GCC "X" Shift Hackathon 2025* (₹25,000 Prize): Built TaskNerve, a multi-tenant AI governance platform delivering data-driven insights and role-based intelligence.
- **Certified in AI/ML:** Completed Machine Learning, Deep Learning, and Python certifications through Naan Mudhalvan and Udemy (2024).
- **International Exposure:** Completed a 4-month mobility program at **INTI, Malaysia** (B.Sc. Computer Science (Hons))
- **AI Certification:** Generative AI certification from Google Cloud (Simplilearn, 2024).

Education

Knowledge Institute of Technology, B. Tech in Artificial Intelligence and Data Science May 2022 – May 2026

- **CGPA:** 8.1/10.0

Experience

- AI Engineer**, Ausweg Info Control Pvt Ltd., Jul 2025– Jan 2026
- Designed and deployed an enterprise LLM-powered EMS chatbot using a full RAG pipeline (chunking, embeddings, vector retrieval, response generation), improving response efficiency by 65%.
 - Built embedding pipelines and semantic similarity search using FAISS, increasing answer accuracy by 72%.
 - Designed and optimized PostgreSQL data models for real-time contextual retrieval integrating EMS APIs and IoT telemetry, reducing lookup time by 60%.
 - Optimized inference performance using quantization, caching, and batching, reducing latency by 40%.
 - Owned end-to-end ML pipeline from data ingestion and preprocessing to deployment and inference optimization in production environments.
- Data Analyst**, Ausweg Info Control Pvt Ltd., Aug 2024– Dec 2024
- Built MQTT-based IIoT data pipelines, improving machine telemetry stability by 38% and reducing communication failures by 27%.
 - Reduced data ingestion latency by 25% through optimized Python data-processing workflows.
 - Enhanced real-time production monitoring visibility by 32%, enabling faster operational decisions.
 - Streamlined debugging and device diagnostics, reducing downtime resolution by 22%.

Projects

TaskNerve — Multi-Tenant AI Governance Platform (GCC Hackathon Winner)

Tech: Next.js, Node.js, PostgreSQL (RLS), Supabase, Mapbox, AI Chatbot

Live: <https://task-nerve.vercel.app>

Github: github.com/rgunasree/TaskNerve.git

- Designed a multi-tenant task orchestration platform enabling autonomous execution with centralized governance.
- Implemented role-aware conversational chatbot for leadership insights and decision support.
- Architected secure multi-tenant database design using PostgreSQL Row-Level Security (RLS).
- Enabled governance-layer abstraction where leadership consumes signals instead of raw operational data.
- Demonstrated improved visibility into task execution and reduced coordination overhead during prototype evaluation across multi-tenant workflows.

Resume Matcher — Hybrid Retrieval & Ranking System

Tech: Python, Streamlit, BM25, FAISS, Sentence Transformers

GitHub: github.com/rgunasree/resume-matcher

- Built hybrid retrieval pipeline combining BM25 (keyword search) and FAISS-based vector search for semantic matching.
- Designed similarity-based ranking system for candidate retrieval and scoring.
- Implemented NLP pipelines for structured extraction from unstructured resumes.
- Automated 90% of screening workflow, reducing time-to-shortlist by 75%.

FairAssess.ai — AI-Powered Hiring Bias Detection Platform

Tech: Next.js 16, TypeScript, React18, Tailwind, NLP, Hugging Face, ML

Live: <https://fairassess.vercel.app/>

GitHub: github.com/rgunasree/FairAssess

- Built bias detection engine using transformer-based zero-shot models achieving 87% detection confidence.
- Developed AI rewriting engine reducing discriminatory language by 43%.
- Applied LLM-based inference pipelines for real-time text analysis and classification.
- Deployed production-ready system with Vercel server-less backend + Playwright E2E testing.
- Impact: Increased diverse applicant rates by 52%.

Technologies

Programming: Python, Java, C, SQL

AI & Machine Learning: LLMs, RAG, NLP, Transformers, Deep Learning, Predictive Modeling, Prompt Engineering, Model Fine-tuning, AI Agents (CrewAI)

Frameworks & Libraries: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Sentence Transformers, Hugging Face

Vector Search & IR: FAISS, BM25, ChromaDB

Deployment & MLOps: Docker, FastAPI, REST APIs, Vercel Server-less Functions

Data Engineering: ETL, ELK Stack

Databases: PostgreSQL, MySQL, SQLite

Frontend & Tools: Next.js, React, Streamlit, Tailwind CSS, Git, VS Code, Jupyter Notebook, Google Colab

Exposure: LangChain, LlamaIndex, AWS (SageMaker, Bedrock, S3, EC2)